LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Practice Worksheet: Balancing Chemical Equations

Chapter / Topic: Chemical Reactions and Equations

For teacher use only. Do not distribute to students.

BALANCE THESE

Q1. $H_2 + O_2 \rightarrow H_2O$
Answer: $2H_2 + O_2 \rightarrow 2H_2O$

Q2. $Mg + O_2 \rightarrow MgO$
Answer: $2Mg + O_2 \rightarrow 2MgO$

Q3. $Na + H_2O \rightarrow NaOH + H_2$
Answer: $2Na + 2H_2O \rightarrow 2NaOH + H_2$

Q4. $Al + O_2 \rightarrow Al_2O_3$
Answer: $4Al + 3O_2 \rightarrow 2Al_2O_3$

Q5. $CH_4 + O_2 \rightarrow CO_2 + H_2O$
Answer: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

Q6. $KClO_3 \rightarrow KCl + O_2$
Answer: $2KClO_3 \rightarrow 2KCl + 3O_2$

Q7. $Fe + H_2O \rightarrow Fe_3O_4 + H_2$
Answer: $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$

Q8. $BaCl_2 + Al_2(SO_4)_3 \rightarrow BaSO_4 + AlCl_3$
Answer: $3BaCl_2 + Al_2(SO_4)_3 \rightarrow 3BaSO_4 + 2AlCl_3$

Q9. $Pb(NO_3)_2 \rightarrow PbO + NO_2 + O_2$
Answer: $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$

Q10. $C_6H_{12}O_6 + O_2 \rightarrow CO_2 + H_2O$
Answer: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$

CHALLENGE

Q11. Write the balanced equation, with physical states, for the reaction of zinc with dilute sulphuric acid. Name the gas formed.
Answer: $Zn(s) + H_2SO_4(aq) \rightarrow ZnSO_4(aq) + H_2(g)$. The gas is hydrogen.